

- Introduction
- Course content development
- Quiz/Assignment
- Practice Problem
- Question and answer Discussion

Discrete Mathematics and Group Theory



- Introduction
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Why Discrete Mathematics?

Can we solve this problem?

Or

Can we say the truth value of this problem?

Can you solve this problem?

Q. Prove that if x is even then x^2 is even.

Q. Prove that if x^2 is even then x is even.

There are two restaurant next to each other. One has a sign that says “ Good food is not cheap” and the other has a sign that says “ cheap food is not good”.

Is both the statement are equivalent?

- Introduction
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“If it rains, I drive to school.”
“It rains.”

Can I say, I drive to school

- Introduction
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It is known that

1. It is not sunny this afternoon, and it is colder than yesterday.
2. We will go swimming only if it is sunny.
3. If we do not go swimming, we will play basketball.
4. If we play basketball, we will go home early.

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Can you conclude “we will go home early”?

What is your answer? The conclusion is true or false

- What is the Discrete Mathematics?

Discrete mathematics is the study of mathematical structures that can be considered "discrete"

Objects studied in discrete mathematics include integers, graphs, and statement in.¹ By contrast, discrete mathematics excludes topics in "continuous mathematics" such as real number, calculus, Euclidean geometry.

- Where Its Applications?

1. Application in computer science.
2. Biological Science.
3. Physical Science.

■ Introduction

Application in computer science

- Advanced algorithms & data structures
 - Programming language compilers & interpreters.
 - Computer networks
 - Operating systems
 - Computer architecture
-
- Database management systems
 - Cryptography
 - Error correction codes
 - Graphics & animation algorithms, game engines, *etc....*
 - *I.e., the whole field!*

Welcome to the course

- Syllabus: unit-I

Logic and Relations:

Logic: Propositions and logical Operations, Conditional statements; Predicate Calculus-First order logic, universal and existential quantifiers; Proof Techniques- methods of proof, Mathematical induction. [T1]

Welcome to course

- Syllabus: unit-I

Relation and Diagraphs- Properties of relations, composition of relations, closure operation on relations, equivalence relations and partitions, paths in relation and diagraphs, Operations on relations, Transitive closure and Warshall's Algorithm. Partial ordered sets (POSET), Hasse diagram, External elements of partially ordered sets, Lattices. [T1]

Welcome to course

- Syllabus: unit-II

Combinatorics:

Combinations and permutations, Enumerating combinations and permutations, Principle of Inclusion-Exclusion and Derangements, Generating functions, Recurrence relations, Solutions of recurrence relations.

[T2]

Welcome to course

- Syllabus: unit-III

Graph Theory:

What is a graph? Matrices and Isomorphism, Special Graphs, Paths, Cycles and Trails, Connected Graphs, Bipartite Graphs, Eulerian paths circuits, Hamiltonian paths and circuits, Fleury's algorithm, Planar Graphs. Embedding and Eulers formula, Colouring of Graphs. [T3]

Welcome to course

- Syllabus: unit-IV

Trees and Networks:

Trees-Basic Properties, Depth first search (DFS) and Breadth first search (BFS) Algorithms, Minimal spanning tree, Shortest path problem, Kruskal's Algorithm, Prim's Algorithm, Dijkstra's Algorithm. Network Flow Problem, Ford-Fulkerson Algorithm, Max-Flow-Min-Cut Theorem. [T3]

Welcome to course

- Syllabus: unit-V

Algebraic Structures and Applications: Binary operations, semi-groups and groups, subgroups, cosets, Lagranges's theorem, Product and quotient semi-groups and groups, Normal subgroup, Homomorphism, coding of binary information and error detection, group codes, decoding and error correction. [T1]

Welcome to course

- **Text Books:**

T1. Discrete Mathematical Structures, Bernard Kolman, Robert Busby, Sharon C. Ross, Pearson Education Inc., New Delhi/Prentice Hall of India (PHI) Pvt. Ltd. New Delhi, Sixth Edition, 2009.

Chapters: 2(2.1-2.4), 4(4.1-4.5, 4.7-4.8), 6(6.1-6.3), 8(8.2), 9(9.1-9.5), 11(11.1-11.2).

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- **Text Books:**

T2. Discrete Mathematics for Computer Scientists and mathematicians, Joe L. Mott, Abraham Kandel, Theodore P. Baker, Prentice Hall of India (PHI) Pvt. Ltd. New Delhi, Second Edition, 2000.
Chapters: 2(2.2-2.5), 3(3.1-3.6).

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- **Text Books:**

T3. Introduction to Graph Theory, Douglas B. West, Prentice-Hall of India Pvt. Ltd., New Delhi, Second Edition, 2003.

Chapters: 1-Sections: 1.1 (up to and including 1.1.40), 1.2 (Up to and including 1.2.27), 1.3(Up to and including 1.3.6), 2-Sections: 2.1 (Up to and including 2.1.13), 2.3(2.3.1-2.3.8), 4- Sections: 4.1 (4.1.1, 4.1.2, 4.1.7 to 4.1.11), 4.3(Upto Including 4.3.11), 5 – Section: 5.1 (Up to including 5.1.21), 6: Section 6.1 (Up to and including 6.1.13, 6.1.21 to 6.1.24), 7 – Section 7.2 (7.2(Up to Including 7.2.19)).

Welcome to course

- **Reference Books:**

1. Discrete Mathematical Structures , Bernard Kolman, Robert Busby, Sharon C. Ross, Pearson Education Inc., New Delhi. / Prentice Hall of India (PHI) Pvt. Ltd., New Delhi, Sixth Edition, 2009.
2. Discrete Mathematics for Computer Scientists and Mathematicians by Joe L. Mott, Abraham Kandel and Theodore P. Baker, Prentice-Hall of India Private Limited, New Delhi, Second Edition, 2000.
3. Elements of Discrete Mathematics, C. L. Liu and D. Mohaptra, Tata McGraw Hill Education, New Delhi, Third Edition, 2008.

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- **Reference Books:**

4. Discrete and Combinatorial Mathematics, Ralph P. Grimaldi, Pearson Education, New Delhi, Fifth Edition, 2005.
5. Discrete Mathematics, Norman L. Biggs, Oxford University Press, Second Edition, Tenth Impression, 2010..
6. Discrete Mathematics and Applications, Thomas Koshy, Second Edition, Elsevier Publication (India), New Delhi.
7. A First look at Graph Theory, John Clark and Derek Allan Holton, Prentice Hall.

Welcome to course

- **Reference Books:**

8. Mathematical Structure for Computer Science, A modern treatment to Discrete Mathematics, J.L. Gersting, W. H. Freeman and Macmillan (India), Fifth / Sixth Edition (Asian Student Editions), 2008.
9. Discrete Mathematical Structures, D. S. Malik and M. K. Sen, CENGAGE Learning India Pvt. Ltd., New Delhi, First Edition, 2005.
10. Discrete Mathematics, Richard Johnsonbaugh, Pearson Education Inc., New Delhi, Seventh Edition, 2010.
11. Discrete Mathematics with Proof, Eric Gossett, Wiley India, Second Edition, Second Edition, 2010 (Reprint).

Propositions

- A *proposition* is a declarative sentence that is either true or false.

Example-1: Toronto is the capital of Canada.

Example-2: $1 + 0 = 1$

Example-3: The Moon is made of green cheese.

Not propositions, examples

Example-1: Sit down!

Example-2: What time is it?

Example-3: $x + 1 = 2$

Example-4: $x + y = z$

Propositional Logic

- Propositional Variables: $p, q, r, s, \dots$
- The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.

Compound Propositions

Compound Propositions; constructed from logical connectives and other propositions

Negation $\neg$

Disjunction $\vee$

Conjunction $\wedge$

Exclusive or $\oplus$

Implication $\rightarrow$

Bi conditional $\leftrightarrow$

Truth Value of Compound Proposition (Negation)

- The *negation* of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

Truth Value of Compound Proposition (Conjunction)

- The *conjunction* of propositions p and q is denoted by $p \wedge q$ and has this truth table:

Truth Value of Compound Proposition (Conjunction)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home and it is raining.”

Truth Value of Compound Proposition (Disjunction)

The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

Truth Value of Compound Proposition (Disjunction)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes “I am at home or it is raining.”

Truth Value of Compound Proposition (Exclusive “or”)

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example: If p denotes “I am at home.” and q denotes “I am at college.” then $p \oplus q$ denotes “I am at home or I am at college.”

Truth Value of Compound Proposition (Implication)

If p and q are propositions, then $p \rightarrow q$ is a *conditional statement* or *implication* which is read as “if p , then q ” and has this truth table:

Truth Value of Compound Proposition (Implication)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “If I am at home then it is raining.”

In $p \rightarrow q$, p is the *hypothesis (antecedent or premise)* and q is the *conclusion (or consequence)*.

Different Ways of Expressing $p \rightarrow q$

- *if p , then q*
 - *if p , q*
 - *q unless $\neg p$*
 - *q if p*
 - *q whenever p*
 - *q follows from p*
 - *a necessary condition for p is q*
 - *a sufficient condition for q is p*
- p implies q*
 p only if q
 q when p
 q when p
 p is sufficient for q
 q is necessary for p

Truth Value of Compound Proposition (Biconditional)

If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as “ p if and only if q .” The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

Truth Value of Compound Proposition (Biconditional)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”

Different Ways of Expressing $p \leftrightarrow q$

Some alternative ways “ p if and only if q ” is expressed in English:

- p is necessary and sufficient for q
- if p then q , and conversely
- p iff q

Example Truth Table

Construct a truth table for $(p \vee q) \rightarrow \neg r$

p	q	r	$(p \vee q)$	$\neg r$	$(p \vee q) \rightarrow \neg r$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	F	T	T

Solution of Construct a truth table for $(p \vee q) \rightarrow \neg r$

	1	2	3	4	5
1					
2					
3					
4					
5					
6					
7					
8					
9					

Solve the following problem:

$$(a) (p \vee \neg q) \rightarrow (p \wedge q)$$

$$(b) (p \wedge q) \rightarrow (p \vee q)$$

$$(c) (p \vee \neg q) \wedge (q \vee \neg p)$$

Solution :

Converse, Contrapositive, and Inverse

From $p \rightarrow q$ we can form new conditional statements .

$q \rightarrow p$ is the **converse** of $p \rightarrow q$

$\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$

$\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It raining is a sufficient condition for my not going to town.”

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

What is the truth values of Converse, Contrapositive, and Inverse of a conditional statement.

From $p \rightarrow q$ we can form new conditional statements .

$q \rightarrow p$ is the **converse** of $p \rightarrow q$

$\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$

$\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Tautologies, Contradictions, and Contingencies

A tautology is a proposition which is always true.

Example: $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Tautologies, Contradictions, and Contingencies

A *contradiction* is a proposition which is always false.

Example: $p \wedge \neg p$

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Tautologies, Contradictions, and Contingencies

A *contingency* is a proposition which is neither a tautology nor a contradiction, such as $p \rightarrow \neg p$

p	$\neg p$	$p \rightarrow \neg p$
T	F	F
F	T	T

What about Tautologies, Contradictions, and Contingencies of following statements.

1. $\sim(p \wedge q) \vee q$
2. $p \rightarrow (p \vee q)$
3. $p \wedge (q \wedge \sim p)$

Logically Equivalent

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

Logically Equivalent

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p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws

$$(a) \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$(b) \neg(p \vee q) \equiv \neg p \wedge \neg q$$

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Logical Equivalence Laws

Identity Laws	$p \wedge T \equiv p$	$p \vee F \equiv p$
Domination Laws	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double Negation law	$\neg(\neg p) \equiv p$	
Negation Laws	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
Commutative laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$

Logical Equivalence Laws

Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$\begin{aligned} &(p \vee (q \wedge r)) \\ &\equiv (p \vee q) \wedge (p \vee r) \end{aligned}$	$\begin{aligned} &(p \wedge (q \vee r)) \\ &\equiv (p \wedge q) \vee (p \wedge r) \end{aligned}$
Absorptions laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$

More Logical Equivalences

Logical Equivalences Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Equivalence Proofs

Question :show that $\neg(p \vee (\neg p \wedge q))$ is logically equivalent to $\neg p \wedge \neg q$.

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) \text{ by second De Morgan Law} \\ &\equiv \neg p \wedge (p \vee \neg q) \text{ by the double negation Law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \text{ by Distributive Law} \\ &\equiv F \vee (\neg p \wedge \neg q) \\ &\equiv (\neg p \wedge \neg q) \text{ BY Identity law}\end{aligned}$$

Equivalence Proofs

Question : Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) \\ &\equiv \neg p \vee \neg q \vee p \vee q \quad \text{negation Law} \\ &\equiv (\neg p \vee p) \vee (q \vee \neg q) \\ &\equiv T \vee T \\ &\equiv T\end{aligned}$$

Equivalence Proofs

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$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) \\ &\equiv \neg p \vee \neg q \vee p \vee q \quad \text{negation Law} \\ &\equiv (\neg p \vee p) \vee (q \vee \neg q) \\ &\equiv T \vee T \\ &\equiv T\end{aligned}$$

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

If the intended meaning is $p \vee (q \rightarrow \neg r)$
then parentheses must be used.

Find the truth value of $p \vee q \rightarrow \neg r$

How many rows are there in a truth table with n propositional variables?

ANSWER: 2^n

Thank you